

Digitizing Market Linkages in The Agricultural Sector: Reducing Middlemen Influence and Promoting Fair Trade

Problem Description

The Indian agricultural sector is hindered by inefficiencies, especially for small-scale farmers who face limited market access and exploitative intermediaries. A 2022 report by NITI Aayog highlights that over 85% of small farmers earn less than ₹20,000 annually due to supply chain gaps and lack of direct consumer access. Buyers, too, suffer from fluctuating prices and limited access to fresh farm produce. These challenges demand a streamlined, transparent system to ensure equitable benefits for farmers and consumers.

Problem Validation

- Traditional supply chains are dominated by middlemen, who extract up to 40% of profits. Studies show that digital interventions can improve income by 20-25% for farmers and reduce consumer costs by 10-15%.
- Despite government efforts like eNAM, adoption remains limited due to accessibility and operational barriers.

Existing Solutions and Limitations

Current apps like eMandi and Kisan Suvidha provide market insights but lack features such as real-time price negotiation, personalized dashboards, and seamless logistics optimization. Many apps cater to large-scale farmers, leaving smallholder farmers underserved.

Proposed Solution: Farm2You

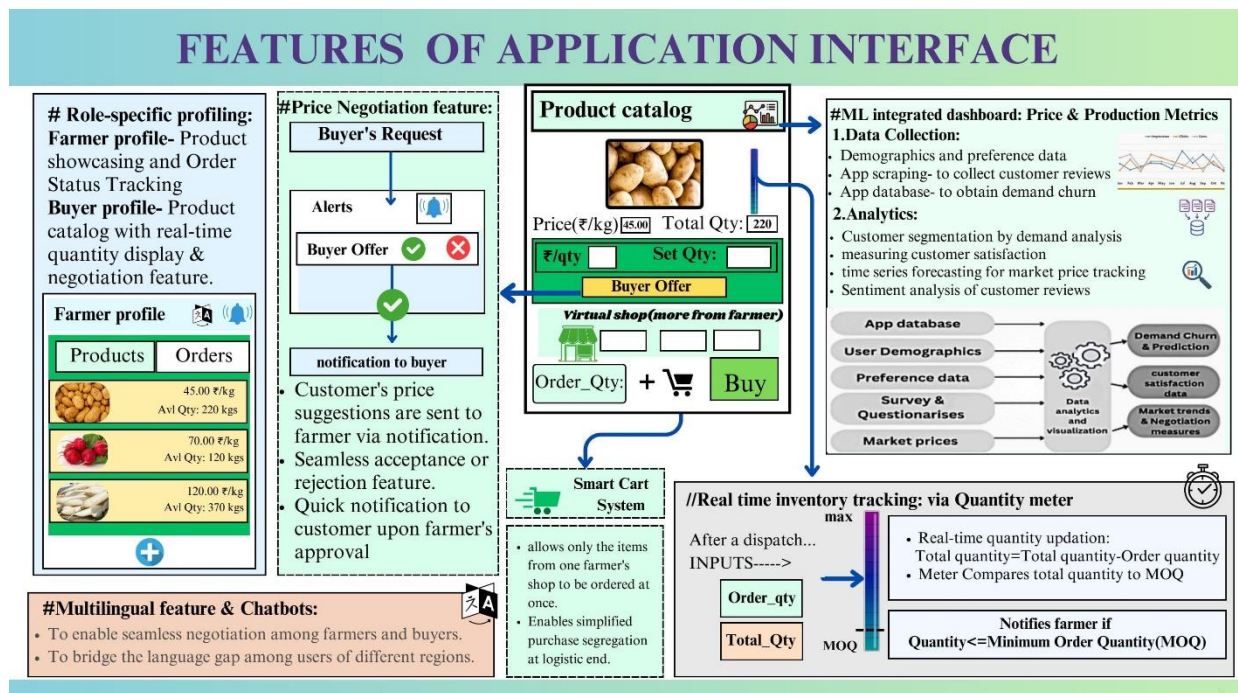
Farm2You connects farmers and buyers directly through a mobile platform, ensuring fair trade, cost-effectiveness, and market transparency. Key features include:

- **Dynamic Product Listing:** Products are ranked by proximity, delivery time, and shelf life.
- **Real-Time Price Negotiation:** Direct communication and price setting between farmers and buyers.
- **Interactive Dashboards:** Data-driven insights using AI/ML for demand forecasting, customer behavior analysis, and production optimization.
- **Logistics Management:** Route optimization and clustering algorithms to minimize delivery costs and carbon footprint.
- **Multilingual Support:** Ensures adoption across diverse demographics.

Technical Feasibility

- The app is built with Kotlin and Jetpack Compose, ensuring smooth UX, while Firebase ensures real-time data synchronization. AI/ML algorithms forecast demand, optimize routes, and analyze user behavior.
- Google Maps API calculates delivery routes, and payment gateways like Razorpay ensure secure transactions.

Prototype



WORKING PROTOTYPE



Kotlin
 Jetpack compose
 Python libraries & ML
 MATPLOTLIB
 Google map API
 Firebase, Firestore
 Chat SDKs
 Mobile Payment Gateway
 Room Database with Kotlin

Uniqueness

Farm2You integrates real-time dashboards, machine learning, and a unique negotiation interface, setting it apart from existing solutions. Its regionalized dispatch system ensures scalability, while data analytics empower farmers to make informed decisions. The platform promotes sustainability, reducing food wastage and environmental footprints.

Conclusion

Farm2You transforms the agricultural sector by digitizing market linkages, improving farmer earnings, and delivering fresh produce to consumers efficiently. By bridging technological gaps, the platform aligns with India's vision of a digitally empowered rural economy.